Compact Visual–Temporal Fusion for RGB-Free Activity Recognition: CUHK-X Challenge 2026 Technical Report

CUHK-X Challenge 2026 · Small Model Track

Technical report · Public version 9 November 2026

Code: https://github.com/EterHF/CUHKX-Small_Track-HAR

Team Information

Field	Value
Team / Kaggle team name	YE Mao1023
Track	Small Model Track
Team members	YE Mao
Affiliation	Peking University
Faculty advisor	Xiujun Ma, School of Artificial Intelligence, Peking University; advisor, not a team member.
Final Kaggle submission	56036959
Observed accuracy	Public 97.512%; private 96.568%

1. One-Paragraph Summary

We address 40-class activity recognition from non-RGB CUHK-X sensor recordings [1]. The final system combines a four-channel R(2+1)D-34 visual classifier operating on colorized depth and infrared (IR) with a skeleton branch comprising a small DSTFormer and a residual temporal convolutional network (TCN). Confidence-weighted late fusion combines their clip-level logits. All inference weights, including YOLO11n, occupy 50.73 MB. Submission 56036959 achieved 97.512% public and 96.568% private leaderboard accuracy, and an offline replay reproduced all 405 submitted predictions.

2. Method Overview

Figure 1 shows two independently sampled branches that meet only at clip-level late fusion.

1. Read the supplied manifest in order and resolve each sample directory without using its identifier as a feature.

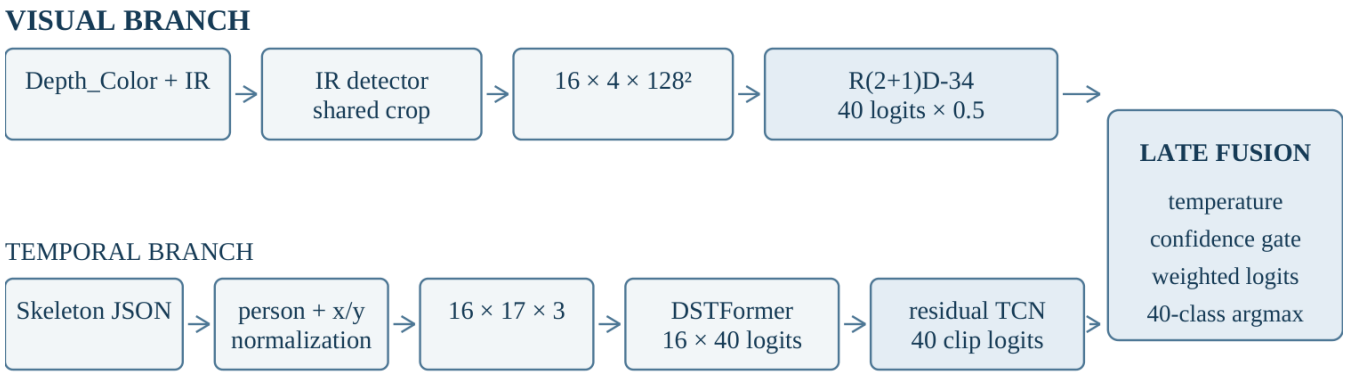
2. Detect a person in up to eight uniformly sampled IR frames with YOLO11n at input size 640 and confidence threshold 0.25. Form one shared union crop from the highest-confidence detection in each probe.

3. Independently sample 16 Depth_Color and 16 IR frames, apply the shared crop, resize to 128 × 128, combine three depth channels with one IR channel, and apply the stored affine transform.

4. Run the four-channel R(2+1)D-34 visual classifier and scale its 40 logits by 0.5.

5. Select 16 skeleton frames. In each frame, choose one 17-joint person and normalize its x–y coordinates. DSTFormer produces frame logits, and a residual TCN converts them to 40 clip logits.

6. Apply fixed temperatures and confidence-dependent branch weights, then take the argmax of the fused logits.



Two independently sampled branches meet only at clip-level logits; all deployed parameters are frozen.

Figure 1. Final visual and temporal branches. Their inputs are sampled independently and combined only through clip-level logits.

If the IR probes yield no person detection above 0.25, each available Depth_Color and IR frame is resized directly to 128 × 128 without cropping.

3. Modalities and Model Design

3.1 Modalities Used

Table 1. Inputs used by the final model.

Modality	Used?	Representation and role
Depth	Yes	Three-channel Depth_Color images; body shape and scene context.
Infrared	Yes	One grayscale channel; person localization and complementary appearance.
Skeleton	Yes	16 × 17 × 3 joint arrays; pose and temporal dynamics.
IMU	No	—
mmWave	No	—
Thermal	No	—

3.2 Visual Branch

The visual branch is a four-channel R(2+1)D-34 [2] with a 40-class head. Its public K-KUNO checkpoint was configured for Depth_Color plus IR and trained on CUHK-X [4], after initialization from IG-65M/Kinetics-400 weights [5]. We selected one ensemble member, kept its architecture, froze the backbone parameters, and optimized the existing 20,520-parameter classifier head for one epoch on the 3,036 official training clips. Backbone BatchNorm statistics also updated. The adapted member was stored with signed per-output-channel 5-bit packing. At inference, weights are decoded to floating point and the branch emits 40 logits scaled by 0.5.

3.3 Temporal Branch

The temporal branch uses a compact task-specific DSTFormer based on MotionBERT [3]. It has width 64, depth 1, two attention heads, 17 joints and a 2,048-unit classification head. It maps each of 16 skeleton frames to 40 logits.

The residual TCN applies LayerNorm(40), convolutions $40 \to 96 \to 96 \to 40$ with kernels 3/3/1 and dilations 1/2/1, GELU and dropout 0.15. Its time-averaged output passes through LayerNorm(40) and Linear(40,40), then is added to the mean of the original frame logits. The decoder has 45,040 parameters.

3.4 Confidence-Weighted Late Fusion

Let r_V and r_T be the visual and temporal logits. We first apply the visual scale and each branch temperature τ_b:

$$z_V = 0.5 r_V, \quad z_T = r_T, \quad s_{b,k} = z_{b,k} / \tau_b \tag{1}$$

Softmax supplies confidence c_b, which is mapped from the 40-class uniform baseline to [0.25,1]:

$$p_b = \mathrm{softmax}(s_b), \quad c_b = \max_k p_{b,k} \tag{2}$$

$$q_b = 0.25 + 0.75 \times \mathrm{clip}((c_b - 1/40) / (1 - 1/40), 0, 1) \tag{3}$$

With base weights $w_V = 0.22$ and $w_T = 0.67$, the effective weights are

$$a_b = w_b q_b / (w_V q_V + w_T q_T) \tag{4}$$

and $a_V + a_T = 1$. The prediction is

$$f_k = a_V s_{V,k} + a_T s_{T,k}, \quad \text{prediction} = \mathrm{argmax}_k f_k \tag{5}$$

The fixed temperatures stored in the submitted bundle are $\tau_V = 1.5271351958543506$ and $\tau_T = 1.1651778344146086$. Only q and a vary by clip.

4. Data Sources

Table 2. Sources that determine the final inputs or weights.

Source	Role	Use in final system
Official challenge data [1]	3,036 labeled clips from users 1–9 and 16–24; anonymous test inputs.	Downstream adaptation, calibration and inference.
K-KUNO checkpoint [4]	Four-channel, 40-class R(2+1)D-34 trained on CUHK-X; initialized from IG-65M/Kinetics-400 [5].	Visual initialization; local classifier-head adaptation and BatchNorm-statistics updates.
CUHK-X DSTFormer [7]	CUHK-X-trained epoch-97 frame classifier.	Frozen temporal initialization at inference.
YOLO11n [6]	COCO-pretrained person detector.	IR person cropping; no local detector training.

5. Training and Validation

5.1 Preprocessing

Within each stream, filenames are ordered and 16 rank-uniform frames are selected. Short streams repeat indices; empty streams yield zero frames. Visual values are converted to [0,1] and mapped to the pretrained backbone's target scale:

$$x' = ((x - \mu_{train}) / \sigma_{train}) \times \sigma_{target} + \mu_{target} \quad (6)$$

The deployed train-derived means are (0.542918655, 0.520943937, 0.245127824, 0.168240735) and standard deviations are (0.417928676, 0.408311983, 0.363606899, 0.185198132) for Depth R/G/B and IR. Target depth means are (0.43216, 0.394666, 0.37645), target standard deviations are (0.22803, 0.22145, 0.216989), and each IR target constant is the arithmetic mean of the three depth constants. These values use only the 3,036 labeled training clips. Sampled images that raise OSError or ValueError during cache construction become zero frames while retaining their temporal positions.

For each skeleton JSON, the person with the highest mean keypoint score is selected. Joints with nonzero third coordinate define the x–y bounding box. Coordinates are centered on that box and scaled by its longer side, preserving proportions; all three components are clipped to [−1,1]. Invalid frames become zeros. The third coordinate is retained as supplied rather than reinterpreted as confidence.

5.2 Downstream Training and Calibration

Table 3. Retained training evidence and downstream reconstruction settings.

Component	Procedure	Recorded boundary
Visual head	Recorded head-only update (20,520 parameters), one epoch; batch 16 inferred from retained BatchNorm counters. Reconstruction uses AdamW, learning rate 0.00002, weight decay 0.01, cosine schedule, smoothing 0.02, flips and seed 2026. Backbone weights are frozen; BatchNorm statistics update.	Two reconstruction runs were byte-identical and reproduced 405/405 fused predictions. They differ from the historical checkpoint bytes; the historical learning rate and complete runtime state were not retained.
DSTFormer	Public epoch-97 checkpoint; no local weight update.	Inherited parameters are preserved in the private final bundle; source information is listed in Section 4.
Residual TCN	Historical five-epoch run; the saved statistic matches epoch 4 full-data training accuracy. Reconstruction uses AdamW, learning rate 0.0003, batch 128, weight decay 0.01, smoothing 0.01, reversal probability 0.25, noise 0.01 and seed 2026.	The reconstruction matches all 45,040 historical parameters. The original invocation is incomplete, and the historical checkpoint has no epoch field.
Calibration	Temperatures selected from 81 log-spaced values in [0.25,4]; fixed branch weights selected on development folds.	Frozen values in Equations (1)–(5) reproduce the archived predictions.

Historical raw-input replay and downstream reconstruction with the retained bootstrap artifacts reproduced all 405 final predictions. Exact checkpoint bytes can vary with the historical runtime state.

5.3 Validation Strategy and Results

The downstream folds hold out users A: 1/6/17/22, B: 2/7/18/23, C: 3/8/19/24, D: 4/9/20 and E: 5/16/21. Fold results guided model selection and are reported as development diagnostics.

Table 4. Validation evidence requested by the official report template.

Requested item	Evidence and interpretation
Validation split	Five downstream cross-subject folds.
Held-out environments	No independent environment holdout.
Local result	2,893/3,036 clips (95.2899%) in the historical flip-TTA development diagnostic. Single-view results are in Section 6.
Public / private leaderboard	97.512% / 96.568% for submission 56036959.
Local/public gap	+2.2221 percentage points; descriptive because protocols differ.
Cross-subject generalization	No separate estimate beyond the development folds.
Expected unseen subject/environment	Unmeasured beyond the private leaderboard; no calibrated forecast.
Deployment variation	Camera and detector changes, pose errors, missing inputs and changed action context.

6. Component Contributions and Storage Precision

These development comparisons use the 3,036 official training clips and the same five downstream user folds. No new leaderboard submission or anonymous-test label was used. For the following accuracy table, each visual model performs a single forward pass without flip test-time augmentation (TTA), matching the deployed visual inference rule. Accuracy uses fold-specific weights; latency uses the final full-training-data bundle with the same architecture and compute rule. Historical flip-TTA diagnostics are separate.

Table 5. Single-view component removal with fixed released fusion settings and 5-bit visual storage.

Configuration	Correct / 3,036	Accuracy
Visual only	2747	90.4809%
Skeleton: frame mean	2845	93.7088%
Skeleton: mean + residual TCN	2854	94.0053%
Visual + skeleton mean	2883	94.9605%
Visual + skeleton + residual TCN	2889	95.1581%

Adding the skeleton branch to the visual branch changes accuracy by +4.677 percentage points (pp); adding the visual branch to skeleton + TCN changes it by +1.153 pp. Their descriptive 95% user-cluster bootstrap intervals are [-0.262, +10.466] and [+0.566, +1.815] pp, respectively, using 10,000 resamples of 18 users. These are downstream development comparisons.

The residual TCN learns a correction to the strong frame-logit mean: output = mean(frame logits) + learned correction. In a separate fixed-budget study (five folds, three seeds, five epochs, final checkpoint), the mean alone reaches 93.7088% and the residual TCN reaches 93.7967% on average, a +0.0878 pp change with interval [-0.209, +0.333] pp. Removing the mean skip connection reaches only 40.1954% under that budget. This supports preserving the pretrained output during short adaptation; it does not establish a comparable gain from temporal modeling or a disadvantage after full convergence. Shuffling frame order yields the same per-seed accuracy. The TCN has 45,040 parameters, compared with 2,456,365 for DSTFormer and 63,514,700 for the visual model.

Table 6. Storage-only quantization from the same source weights and bundle container.

Visual storage	Full bundle (MB)	Single-view fused accuracy
FP16	137.9074	95.1910%
5-bit	50.7151	95.1581%

The 5-bit bundle is 63.23% smaller. Single-view fused predictions change on 1 of 3,036 clips (net correct change: -1). The table uses newly serialized containers; the submitted artifact remains 50.733469 MB. Both formats decode to FP32 resident parameters and use identical floating-point kernels. Therefore, 5-bit packing is a storage benefit, not native 5-bit execution.

7. Inference Efficiency and Deployment Reference

7.1 Measurement Protocol

Measurements were taken on 10 October 2026 using an Intel Xeon Platinum 8368Q at 2.60 GHz (152 logical CPUs, 1 TiB host RAM), four PyTorch CPU threads, and one NVIDIA A100-SXM4-40GB (physical GPU 4) on a shared server. Python is 3.11.15 and PyTorch is 2.13.0+cu129. CPU execution uses FP32; the GPU uses FP16 autocast for visual inference and BF16 autocast for skeleton/TCN. Deterministic cuDNN is enabled. These are server reference measurements; mobile/edge hardware was not measured.

Before each of six GPU workers, the target GPU reported 0–0 MiB in use. During GPU measurements, whole-host one-minute load average ranged from 4.77 to 5.93; during CPU measurements it ranged from 5.76 to 12.60. GPU utilization includes the benchmark itself. CPU affinity and GPU clocks were not locked, and the shared host was not isolated.

FP16 and 5-bit run in separate fresh processes in order FP16, 5-bit, 5-bit, FP16. Each core/prepared condition pools two rounds: CPU B1 uses 3 warmups and 15 measurements per round; GPU B1 and B16 use 10 warmups and 50 measurements. CUDA is synchronized around wall-clock timers. Core timing includes both classifiers with device-resident inputs; prepared timing adds host-input transfers, frame-logit CPU staging and the CPU fusion gate. Neither includes detector or file I/O. Small differences between storage formats are observational and do not establish a speedup.

Raw-pipeline measurements use 80 distinct training clips: two clips from different users for each of 40 classes, with user counts balanced greedily before timing, covering all 18 users. Each

format/device uses a fresh process, two warmup clips and one timed pass. This class-covered follow-up replaces an initial 18-clip timing sample that covered only two classes; both sets remain archived and are not pooled. Timing includes raw file I/O, eight IR probes, YOLO11n, frame preprocessing, all classifiers, transfers and fusion in a persistent process. Model loading, cache/CSV writes and full-dataset CLI orchestration are excluded; filesystem caches are uncontrolled and warm. Percentiles from this small sample remain descriptive.

Table 7. Warm single-clip latency. B1 denotes batch size one; p50 / p95 are median / 95th percentile.

Device / storage	Core B1 p50 / p95 (ms)	Prepared B1 p50 / p95 (ms)	Raw B1 p50 / p95 (ms)
CPU / FP16	664.35 / 677.31	666.73 / 720.65	1121.90 / 1185.07
CPU / 5-bit	700.00 / 715.09	701.52 / 710.40	1088.17 / 1141.87
A100 / FP16	9.72 / 10.13	10.35 / 10.51	181.35 / 226.68
A100 / 5-bit	9.45 / 9.58	10.15 / 10.55	180.24 / 224.03

Table 8. Additional GPU throughput reference, including transfers and fusion.

Storage	Prepared B16 p50 / p95 (ms)	Clips / second	GPU alloc. / reserved (GiB)
FP16	66.14 / 67.65	238.98	1.567 / 2.135
5-bit	66.05 / 67.64	239.06	1.567 / 2.135

7.2 Memory and Loading

Table 9. Raw-pipeline B1 memory including the detector, measured in separate processes without preceding batch benchmarks. Host RSS reports the sampled raw-phase peak / whole-process high-water mark including loading and warmup. GPU measurements cover the raw phase.

Device / storage	Host RSS raw / total (GiB)	Torch GPU alloc. / reserved (GiB)	Sampled process GPU (GiB)
CPU / FP16	1.666 / 1.696	—	—
CPU / 5-bit	1.574 / 1.574	—	—
A100 / FP16	2.090 / 2.091	0.448 / 0.514	1.043
A100 / 5-bit	2.117 / 2.117	0.448 / 0.514	1.043

Resident classifier parameters occupy 251.83 MiB in either format, before activations, workspaces, detector, runtime and allocator overhead. PyTorch allocated memory is live tensor/workspace storage; reserved memory includes its caching allocator. The process GPU measurement also includes CUDA/library overhead. Raw-phase host RSS is sampled every 50 ms and process GPU memory

about once per second, so short peaks may be missed; the whole-process host high-water mark uses the operating-system counter. None of these values is a claimed minimum device capacity or the peak RAM of the full-dataset inference CLI. MB is decimal; MiB/GiB are binary.

Table 10. Mean loading-stage sum over two fresh processes. Imports, detector startup and untimed garbage-collection gaps are excluded; this is not total CLI startup wall time. OS page cache is not flushed.

Storage	CPU decode (s)	CPU model ready (s)	A100 model ready (s)
FP16	0.035	0.913	1.355
5-bit	3.466	4.293	4.667

The smaller 5-bit file adds a one-time unpack/dequantization cost. It provides no resident-parameter memory reduction in the released loader. Keeping models resident amortizes loading. The underlying samples, round medians, machine-load observations and module timings are retained privately and are not part of the public source release.

7.3 Cost of Each Branch

Table 11. Isolated component timing with all classifier weights resident, 5-bit storage and B1.

5-bit storage, B1 core	CPU p50 (ms)	A100 p50 (ms)
Visual backbone + head	698.053	6.801
DSTFormer + mean	3.101	2.510
DSTFormer + residual TCN	3.542	2.972
Residual TCN alone	0.411	0.463
Both branches, mean	700.928	9.021
Both branches, residual TCN	699.995	9.454

Separate medians are not additive. The skeleton path adds relatively few parameters, while the 45,040-parameter residual is only 0.0682% of classifier parameters. Its low cost supports retaining it as a small correction, but the accuracy study does not establish a strong independent temporal-order benefit. Visual inference is the principal classifier cost; raw preprocessing also limits end-to-end GPU throughput.

8. Public Release and Reproducibility

The source release at https://github.com/EterHF/CUHKX-Small_Track-HAR provides training and inference code, configuration and this report. Model checkpoints, competition data, caches and archived predictions are excluded; the competition artifacts were supplied separately for organizer verification.

Setup instructions accompany the release. Project-authored code is licensed under Apache 2.0; copyright and license notices are included in NOTICE. Third-party components retain their applicable licenses. Original report text and figures are licensed under CC BY 4.0: https://creativecommons.org/licenses/by/4.0/ .

9. References

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